

Beyond Average Accuracy: Statistical Distinguishability and Temporal Concentration in Data-Rich Macroeconomic Forecasting

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Abstract

Machine-learning forecasts can have lower average errors than simple benchmarks without establishing a statistically distinguishable or persistent advantage. This study examines these distinctions in a fixed-length rolling pseudo-out-of-sample exercise based on a July 2026 current-vintage FRED-MD snapshot. Four U.S. monthly targets - CPI inflation, PCE inflation, industrial production growth, and the change in unemployment - are forecast at horizons of 1, 3, 6, and 12 months. The comparison covers nine individual models and three forecast combinations, using 360-month estimation windows and up to 90 target months from December 2018 to May 2026. XGBoost records the lowest average relative RMSE (0.821) against transformed-series persistence, followed closely by the random-forest variants; forecast combinations also deliver sizable numerical gains. Formal inference is much less conclusive. After Holm adjustment, none of the 176 squared-error Diebold-Mariano comparisons rejects equal predictive accuracy in favor of an alternative. Five absolute-error comparisons do so, all at the one-month industrial-production horizon. A 90% Model Confidence Set retains all twelve approaches in 25 of 32 target-horizon-loss panels and never retains fewer than ten. Rankings also vary over time: in 12-month rolling evaluations, winner sets change in 18% to 38% of adjacent windows. Within each model-target-horizon cell, the twelve months making the largest positive and negative contributions account, on average, for 85.6% of gross improvements and 89.1% of gross deteriorations relative to the benchmark. These patterns remain under longer ranking windows, and the average gain falls sharply when the benchmark's largest-error dates are removed. Lower average loss, statistical distinguishability, and temporal durability are therefore separate properties. For this short, shock-heavy sample, the evidence supports reporting forecast combinations and uncertainty diagnostics alongside the leading models rather than treating one ex post winner as definitive.

Keywords: macroeconomic forecasting; machine learning; FRED-MD; Diebold-Mariano test; Model Confidence Set; forecast stability

JEL classification: C52, C53, C55, E37

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1 Introduction

High-dimensional statistical learning is now a standard part of macroeconomic forecasting. Monthly panels contain many potentially informative series, but standard low-dimensional autoregressions cannot use them all without dimension reduction or regularization. Penalized regressions, factor models, random forests, and boosting address this problem in different ways and have generated meaningful gains for selected variables and horizons (Stock and Watson, 2002a, 2002b; Medeiros et al., 2021; Goulet Coulombe et al., 2022). More recent studies examine recurrent and Bayesian neural networks, large language models, time-series foundation models, and alternative data transformations (Daniele et al., 2024; Paranhos, 2025; Hauzenberger et al., 2025; Carriero et al., 2025, 2026).

Many empirical comparisons nevertheless begin and end with average RMSE or MAE. That is not enough for a model-selection decision. An observed gain may be too imprecise to distinguish from sampling variation, may disappear for another target or horizon, or may be generated by a small number of influential months. The last concern is especially important when the evaluation sample contains the COVID-19 collapse and reopening, because a few observations can dominate squared-error comparisons.

The analysis below examines average accuracy, inferential uncertainty, and temporal durability in a common forecasting experiment. It uses a current-vintage FRED-MD panel, four representative U.S. monthly targets, four direct forecast horizons, a fixed-length 360-month rolling estimation window, and 90 pseudo-out-of-sample target months before target-specific exclusions. The model universe contains nine individual approaches - transformed-series persistence, AR(4), Ridge, LASSO, Elastic Net, PCA factor regression, Random Forest, XGBoost, and Boruta RF - together with mean, median, and leakage-safe inverse-RMSE combinations. Every approach is evaluated with the same origin-specific information set, predictor eligibility rules, lag construction, missing-data protocol, and target dates.

Rather than proposing a new forecast-comparison statistic, the analysis integrates three forms of evidence from the same archived forecasts. First, accuracy rankings are reported alongside pairwise Diebold-Mariano tests against the persistence benchmark and a 90% Model Confidence Set, so numerical leadership is not equated with statistically distinguishable superiority. Second, temporal durability is examined through rolling winner turnover, date-level concentration of gross improvements and deteriorations, and benchmark-tail attribution. Third, the operational design is made explicit: target-specific transformations, causal gap filling, a rolling missing-data protocol with a PCA-EM fallback, feature-selection fallbacks, leakage-safe ensemble weights, and a computational release tied to a fixed data checksum.

The numerical and inferential conclusions differ sharply. XGBoost has the best average RMSE rank and lowers RMSE by about 17.9% relative to the benchmark across the sixteen target-horizon cells. Random forests and forecast combinations are also competitive. Yet none of the squared-error Diebold-Mariano comparisons is significant after multiplicity correction. Under absolute-error loss, significant gains appear only in five one-month industrial-production comparisons. The Model Confidence Set retains every approach in 25 of 32 target-horizon-loss panels and retains at least ten in every panel. A seemingly decisive ranking therefore coexists with broad statistical uncertainty.

The timing of the errors points in the same direction. In 12-month rolling evaluations, the RMSE winner changes 14 to 30 times across 76 or 78 adjacent-window transitions, depending on the target and horizon. Within each model-target-horizon cell, the twelve largest positive monthly contributions account for about 86% of gross improvement on average, while the twelve largest negative contributions account for about 89% of gross deterioration. Removing only the date with the largest RW squared error in each target-horizon cell raises the best average relative RMSE from 0.821 to 0.916; after twelve such dates are removed, RW leads on average. The numerical ranking is therefore sensitive to a small group of difficult months.

These results do not imply that machine learning is unhelpful. They instead show why three questions should be kept separate: which model has the lowest average loss, which models can be statistically distinguished, and whether the measured advantage persists through time. A short, shock-heavy out-of-sample period provides much less certainty when these questions are answered separately than when the ranking alone is reported.

Section 2 places the analysis in the data-rich forecasting and forecast-evaluation literatures. Section 3 describes the data, transformations, missing-data protocol, and pseudo-out-of-sample design. Sections 4 and 5 present the forecasting models and evaluation methods. Section 6 reports the empirical results, Section 7 documents implementation diagnostics and reproducibility, and Sections 8 and 9 discuss limitations and conclusions.

2 Related Literature

2.1 Data-rich macroeconomic forecasting

The modern data-rich forecasting literature originates in the use of diffusion indexes and principal components to summarize large predictor panels. Stock and Watson (2002a, 2002b) show how a small set of estimated factors can improve forecasts when many macroeconomic indicators are available. McCracken and Ng (2016) provide FRED-MD, a monthly database with transformation codes and regularly updated coverage that has become a common benchmark for high-dimensional U.S. macroeconomic forecasting.

Penalized regression and machine-learning methods extend factor forecasting by imposing different forms of regularization and nonlinearity. Ridge, LASSO, and elastic net trade bias for variance reduction in high-dimensional designs (Hoerl and Kennard, 1970; Tibshirani, 1996; Zou and Hastie, 2005). Random forests and gradient boosting introduce nonlinear partitions and interactions (Breiman, 2001; Friedman, 2001; Chen and Guestrin, 2016). Medeiros et al. (2021) show that machine-learning methods can improve inflation forecasts in a data-rich environment, while Goulet Coulombe et al. (2022) decompose the sources of machine-learning gains and emphasize the role of nonlinearities, regularization, validation, and loss functions. Related evidence shows that transformations themselves materially affect the implicit regularization of machine-learning forecasts (Goulet Coulombe et al., 2021). Masini et al. (2023) provide a broader review of machine-learning advances for time-series forecasting.

Neural methods form an important but methodologically distinct branch of this literature. Paranhos (2025) finds that LSTM forecasts of U.S. inflation are competitive, particularly at longer horizons and in uncertain periods, while also documenting sensitivity to network architecture and initialization. Hauzenberger et al. (2025) design Bayesian neural networks for the small-T, large-K structure of macroeconomic datasets by combining shrinkage and stochastic volatility. Work on large language models and time-series foundation models adds another design problem: external training data, temporal contamination, and vintage consistency become part of the forecasting protocol rather than secondary implementation details (Carriero et al., 2025, 2026).

Kotchoni et al. (2019) provide the closest broad antecedent. They compare 31 forecasting approaches from six model classes over a long 1970-2014 pseudo-out-of-sample period, report forecast combinations and Model Confidence Sets, and study the stability of forecast performance. The present paper does not claim that model uncertainty or time variation in forecast rankings is new. Its narrower contribution is to connect multiplicity-adjusted benchmark tests, multi-model confidence sets, rolling winner turnover, date-level loss concentration, and benchmark-tail attribution in a post-2018, shock-heavy monthly FRED-MD application that includes tree-based machine learning and an origin-specific, fully documented preprocessing protocol.

Accordingly, the exercise does not seek a universally superior algorithm. Instead, it asks how much confidence can be placed in performance differences generated by a common and operationally explicit pipeline. This focus complements studies centered on new architectures, model interpretation, or genuine real-time vintages.

2.2 Forecast comparison and model uncertainty

The Diebold-Mariano test evaluates the null of equal expected predictive loss for two forecast sequences (Diebold and Mariano, 1995). The test is widely used, but its interpretation depends on the loss function, serial dependence, sample length, and the research design; small samples and model selection can materially reduce power (Diebold, 2015). Because this paper compares eleven alternatives with a common benchmark at each target and horizon, raw pairwise p-values are adjusted using Holm's sequentially rejective procedure (Holm, 1979).

Pairwise tests do not directly summarize uncertainty across the full model set. Hansen et al. (2011) address that problem with the Model Confidence Set, which sequentially removes inferior models while controlling the confidence level. A large survivor set is not evidence that every model is equally accurate; it means the available loss sample is insufficient to eliminate most candidates at the chosen confidence level. This distinction is central to the results below.

2.3 Temporal stability and concentration

Average forecast losses can conceal substantial instability. Giacomini and Rossi (2010) develop formal methods for local forecast comparison in unstable environments, and Rossi (2013) reviews the broader literature. Kotchoni et al. (2019) document variation in forecast performance across time and economic conditions. Siliverstovs and Wochner (2021) show, using a large quarterly U.S. panel, that full-sample rankings can be disproportionately shaped by recessions and use time-specific contributions to trace that state dependence. Related work associates nonlinear forecasting gains with uncertainty and financial frictions, while the COVID episode illustrates how extreme observations can dominate empirical comparisons (Goulet Coulombe et al., 2022; Lenza and Primiceri, 2022). The diagnostics used below are descriptive rather than substitutes for formal instability or state-dependence tests: rolling winner turnover records changes in the leading set, and loss concentration measures whether an average advantage is broadly distributed or concentrated in a few dates.

Forecast instability also provides a classical rationale for forecast combination. Bates and Granger (1969) show that combining imperfect forecasts can improve accuracy when their errors contain complementary information, and Timmermann (2006) reviews why simple combinations often remain competitive when the identity of the best model changes over time. The combinations in this paper are therefore not treated as an afterthought; they are a direct response to model uncertainty and ranking instability.

Building on these literatures, the analysis evaluates average loss, inferential uncertainty, and temporal durability separately within a fixed information set and a prespecified model universe.

3 Data and Experimental Design

3.1 FRED-MD panel and target variables

The data are drawn from a current-vintage FRED-MD monthly snapshot frozen in July 2026. The raw file contains 126 economic series plus the date field, covering January 1959 through May 2026. New orders for consumer goods (ACOGNO) fail the common initial-window availability criterion and are excluded. Each target-specific experiment therefore begins with 125 transformed series: the selected target and 124 candidate predictors. The checksum identifying the frozen data snapshot is reported in Appendix B.

Before model construction, each series is transformed according to its series-specific FRED-MD transformation code. These transformations are designed to remove deterministic or stochastic trends, stabilize scale, and improve the stationarity of the resulting series; they are not formal stationarity tests and do not guarantee covariance stationarity in every rolling window. The seven codes are 1: x_t ; 2: first difference of x_t ; 3: second difference of x_t ; 4: $\log x_t$; 5: first difference of $\log x_t$; 6: second difference of $\log x_t$; and 7: first difference of $(x_t/x_{t-1} - 1)$. Except for the two prespecified price-target overrides described below, every target and predictor retains its official FRED-MD code.

Four targets represent prices, real activity, and the labor market. For CPIAUCSL and PCEPI, the forecast object is defined using transformation code 5, the first difference of log prices. This research-specific override is not a correction to the FRED-MD convention: the database's official code 6, the second difference of log prices, would make the dependent variable the monthly change in inflation (approximately inflation acceleration), whereas code 5 makes it monthly inflation itself. The latter definition matches the economic object studied here and keeps the cumulative-change and reconstructed-level interpretations transparent. INDPRO uses the official first difference of logs, and UNRATE uses the official first difference. Table 1 summarizes the target definitions.

Table 1. Core targets and transformations

Code	Category	Primary transformation	Transformation basis	Start	End
CPIAUCSL	Prices	First difference of log	Prespecified research transformation	1959-01-01	2026-05-01
PCEPI	Prices	First difference of log	Prespecified research transformation	1959-01-01	2026-05-01
INDPRO	Output and Income	First difference of log	Official FRED-MD transformation	1959-01-01	2026-05-01
UNRATE	Labor Market	First difference	Official FRED-MD transformation	1959-01-01	2026-05-01

Notes: CPIAUCSL and PCEPI use a prespecified research transformation (FRED-MD transformation code 5 rather than the database's code 6) so that the primary target is monthly inflation rather than the monthly change in inflation. These overrides define the forecast object for this study and are not corrections to the official FRED-MD codes. INDPRO and UNRATE use their official FRED-MD transformations.

3.2 Fixed-length rolling-window pseudo-out-of-sample design

The evaluation is a current-vintage pseudo-out-of-sample exercise rather than a genuine real-time-vintage experiment. At each forecast origin, models use only observations dated at or before that origin, but the historical series are taken from a single current-vintage FRED-MD file. This prevents direct look-ahead within the rolling algorithm but does not reconstruct the data releases that were actually available in real time. Robustness to historical data vintages and release lags is therefore left for subsequent work.

For each target, a fixed-length 360-month rolling window is used to estimate direct forecasts at horizons $h=1, 3, 6, 12$ months. The OOS target-month range is December 2018 through May 2026, yielding 90 target months before target-imputation exclusions. Predictors at origin t are aligned directly with the transformed target at $t+h$. The feature vector uses four monthly input blocks: the value observed at the origin and three monthly lags (lags 0-3). When all predictors are active, the full design contains the target, 124 predictors, and four PCA factors, for a maximum of 129 variables multiplied by four input blocks, or 516 regressors. The factor model uses the transformed target and four factors with between one and four monthly input blocks.

$$y_{t+h} = f_h(z_t, z_{t-1}, z_{t-2}, z_{t-3})$$

where z_t contains the transformed target, the active FRED-MD predictors, and, for the full models, four principal components. At every origin, the components are re-estimated from the eligible and completed predictor panel after centering and scaling each column; the target is excluded from this factor-extraction step. The factor scores are appended to the transformed target and raw transformed predictors in the full design. The factor model instead uses only the transformed target and factor scores. All models are re-estimated at each forecast origin.

3.3 Missing data and rolling-window imputation protocol

Target and predictor missingness are treated separately. For a target, causal last observation carried forward (LOCF) is applied to the raw level series only for internal gaps of no more than three months; leading values and observations beyond the final reported value are not imputed. The target-specific transformation is then applied. Any transformed actual whose construction uses an imputed raw value is flagged and excluded from formal evaluation, and the origin level used for reconstructed-level forecasts must be observed rather than imputed. For a first-difference or first-log-difference target, one imputed raw month can affect that month and the following transformed observation. CPIAUCSL and UNRATE each contain one internally imputed raw month, which removes two transformed target months and reduces the primary monthly evaluation sample to 88 observations. PCEPI and INDPRO retain all 90.

Predictors are transformed before rolling-window screening. Within each 360-month estimation window, a predictor is eligible only if it has at least 24 finite observations, a finite-observation share of at least 90%, and nonzero finite variation. Variables that fail these conditions are excluded from that window, and leading missing blocks are never backfilled. For eligible predictors, gaps - including short ragged-edge gaps - are filled causally by last observation carried forward for at most three consecutive months. Any cells still missing after this step are passed to an iterative four-factor PCA reconstruction used as the final fallback.

For each forecast origin, let W denote the current matrix containing the complete transformed target and the eligible transformed predictors. The target column is never altered by the PCA fallback; any permitted target gap has already been handled by the separate rule above. Historical target values may nevertheless contribute to the low-rank reconstruction used to fill predictor cells. Column means and standard deviations are computed from observed entries and then held fixed. After standardization, missing cells are initialized at zero, which corresponds to the observed-column mean on the original scale. At iteration r , a rank-four PCA approximation is fitted to the current completed matrix. Only cells that were originally missing are replaced by their common-component reconstructions; observed cells remain fixed. This deterministic EM-style reconstruction follows the general incomplete-data logic of Dempster et al. (1977) and is closely related to EM algorithms for PCA with missing information (Roweis, 1997). The factor count is fixed ex ante at four.

$$\delta_r = \sqrt{\frac{1}{|M|} \sum_{(i,t) \in M} (z_{it}^{(r)} - z_{it}^{(r-1)})^2}$$

where M is the set of originally missing cells. Iteration stops when the root-mean-square change δ_r falls below 10^{-4} or after 300 iterations. A window that reaches the iteration limit is retained but explicitly flagged rather than silently classified as converged. In the frozen evaluation, 274 of the 360 rolling windows for each target contained at least one predictor missing value before eligibility screening and short-gap filling. Screening removed ineligible variables, and causal filling resolved the remaining eligible gaps, so iterative PCA reconstruction was never activated: the maximum recorded EM iteration count was zero for all four targets. There were no nonconverged windows and no residual missing values. The complete screening, gap-

filling, factor-extraction, and model-estimation sequence is repeated at every forecast origin using only the current rolling window.

3.4 Primary and secondary evaluation tracks

The primary track evaluates the target-month transformed value. For price and industrial-production targets, this is a monthly log change; for unemployment it is a monthly first difference. The secondary track is estimated as a separate direct horizon-specific response rather than as a recursive sum of one-step forecasts. For CPIAUCSL, PCEPI, and INDPRO, the response is $\log L_{t+h} - \log L_t$, and the reconstructed level is obtained by multiplying L_t by the exponential of the predicted cumulative log change. For UNRATE, the response is $L_{t+h} - L_t$, and the reconstructed level adds the predicted cumulative arithmetic change to L_t . The analysis emphasizes the primary track because it is common across all targets and avoids mixing conceptually different multi-month objects. Secondary results are preserved in the replication bundle.

4 Forecasting Models

The experiment compares nine individual forecasting approaches - two low-complexity benchmarks and seven data-rich models - and three forecast combinations under a common information set. Stochastic learners use prespecified origin-specific seeds, and all tuning and feature selection rely only on the current rolling training sample. Table 1A summarizes the role of each model family.

Table 1A. Forecasting model families and experimental roles

Model family and forecasts	Main mechanism	Role in the experiment
Baselines: RW; AR(4)	Transformed-series persistence and autoregressive target dynamics	Low-complexity benchmarks
Penalized linear: Ridge; LASSO; Elastic Net	L_1/L_2 shrinkage and variable selection	High-dimensional linear regularization
Factor: PCA factor regression	Dimension reduction followed by dynamic regression	Common-component benchmark
Bagged trees: Random Forest	Averaging of decorrelated regression trees	Nonlinearities and interactions
Boosting: XGBoost	Sequential gradient-boosted regression trees	Flexible nonlinear learner
Selected trees: Boruta RF	All-relevant feature screening before random forest	Feature-selection robustness
Combinations: Mean; Median; Inverse-RMSE	Cross-model diversification	Reduction of model-specific instability

The model universe is deliberately bounded. The exclusion of neural networks does not imply that monthly macroeconomic panels are inherently unsuitable for them. A credible comparison within a 360-month rolling window would require a separate protocol for architecture choice, regularization, initialization, stochastic training variation, and time-ordered validation. Pretrained time-series or language models would also introduce external training data and potential temporal or vintage contamination. This experiment therefore concentrates on established linear, factor, and tree-based methods plus forecast combinations, and its claims are limited to that model universe.

4.1 Baselines

RW is a persistence forecast for the transformed target: the h -step-ahead transformed value is set equal to the latest transformed observation available at the forecast origin. It should therefore be interpreted as transformed-series persistence. In the secondary reconstructed-level track, the corresponding no-change level

benchmark has a different economic meaning. AR(4), as labeled in the implementation, is a direct horizon-specific regression on the four target observations available at the origin: y_t , y_{t-1} , y_{t-2} , and y_{t-3} .

4.2 Regularized linear models

Ridge, LASSO, and Elastic Net use the common direct design, which has as many as 516 columns when every predictor and all four monthly input blocks are active. The mixing parameter α is fixed at 0 for Ridge, 1 for LASSO, and 0.5 for Elastic Net. Predictor columns are standardized internally within each glmnet fit, and an intercept is included. For each rolling window, the penalty is selected with a time-ordered validation holdout equal to the larger of twelve observations or 20% of the available direct-training rows. The value of λ minimizing validation mean squared error is chosen, and the model is then refitted on the full rolling training sample.

All three estimators solve a common penalized least-squares problem, with α governing the balance between L_1 and L_2 regularization; the unpenalized intercept is suppressed in the notation:

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{2n} \sum_{t=1}^n (y_t - x_t' \beta)^2 + \lambda \left(\frac{1-\alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1 \right)$$

where Ridge sets $\alpha=0$, LASSO sets $\alpha=1$, and Elastic Net sets $\alpha=0.5$. This common formulation holds the information set and validation protocol fixed while varying the form of shrinkage.

4.3 Factor and tree-based models

The factor model regresses the future transformed target on its own current and lagged values and four principal components extracted from the predictor panel. BIC selects between one and four monthly input blocks, where four blocks mean the origin value plus three lags. Schematically, with the sums taken over the selected input blocks and four factors,

$$y_{t+h} = c_h + \sum_j \phi_{hj} y_{t-j} + \sum_k \sum_j \gamma_{hjk} F_{k,t-j} + \varepsilon_{t+h}$$

where $F_{k,t}$ is the k th principal component and p_h is the BIC-selected number of monthly input blocks. Random Forest provides a nonlinear benchmark with limited tuning by averaging decorrelated trees, whereas XGBoost builds trees sequentially and can adapt more aggressively to interactions and localized nonlinearities. Random Forest uses 600 trees, a minimum terminal-node size of five, and $mtry$ equal to the integer square root of the current feature count. XGBoost uses squared-error loss, a learning rate of 0.05, maximum depth four, subsample 0.90, a column subsampling rate of 0.75, and up to 800 boosting rounds. A time-ordered 20% validation holdout selects the number of rounds, with early stopping after 25 rounds without improvement; the model is then refitted on the full rolling sample for the selected number of rounds.

Boruta RF applies the Boruta all-relevant feature-selection procedure (Kursa and Rudnicki, 2010) to the lagged design features at the first forecast origin and every six origins thereafter. Each refresh allows up to 100 Boruta runs, resolves tentative attributes using the TentativeRoughFix rule, and then fits a 600-tree random forest to the confirmed features. If a refresh confirms no feature, a prespecified variance-ranking fallback selects the ten highest-variance design features. One such fallback occurs for unemployment at the three-month horizon and affects six forecasts. The event is retained and disclosed rather than silently discarded.

4.4 Forecast combinations

Three combinations use the nine individual forecasts. The mean and median ensembles apply equal weighting at each origin. For the inverse-RMSE ensemble, the weight on member m at origin t is

proportional to $RMSE_{m,t}^{-1}$, normalized so that the nine weights sum to one. To avoid using outcomes that would not yet be observable at the forecast origin, $RMSE_{m,t}$ is computed only from forecasts whose target dates are no later than the current origin date. At least twelve realized errors per member are required before inverse-RMSE weights replace equal weights.

5 Evaluation and Statistical Inference

5.1 Accuracy measures

For each target, horizon, and model, forecast errors are summarized by root mean squared error (RMSE), mean absolute error (MAE), bias, and correlation. Relative RMSE and relative MAE divide a model's metric by the corresponding RW metric for the same target, horizon, and matched dates; values below one favor the alternative. DM and MCS calculations use the intersection of valid target dates available for all twelve approaches within a target-horizon-track panel, so every formal comparison is based on an identical sample.

5.2 Diebold-Mariano tests

Pairwise Diebold-Mariano tests compare each alternative with RW under squared-error and absolute-error loss. The implementation uses the Harvey-Leybourne-Newbold finite-sample modification of the Diebold-Mariano statistic (Harvey et al., 1997), as implemented in the R forecast package. Tests are two-sided, the horizon parameter equals the forecast horizon, and the long-run variance uses Bartlett weights. Within each target-horizon-loss panel, the eleven p-values are adjusted by Holm's procedure. The loss differential is defined as alternative loss minus RW loss, so a negative mean favors the alternative. With a short OOS sample and overlapping multi-step errors, failure to reject indicates limited statistical separation rather than exact equality.

5.3 Model Confidence Set

The MCS is estimated separately for each target, horizon, and loss using the T_{max} statistic, 5,000 block-bootstrap replications, and $\alpha=0.10$. The implementation selects block length automatically, subject to a minimum of three observations, using the R package MCS (Bernardi and Catania, 2018); the selected lengths in the reported panels range from 3 to 13. The resulting 90% set contains models that have not been statistically eliminated. Membership is not proof that a model is best; a wide set may reflect similar losses, low power, or both.

5.4 Temporal stability and concentration

Two descriptive diagnostics complement average loss and formal inference. First, RMSE rankings are recomputed in rolling 12-month evaluation windows, and winner-set changes are divided by the number of adjacent-window transitions. Second, gross monthly loss reductions and deteriorations relative to RW are ranked separately within each model-target-horizon cell. The share attributable to the twelve largest contributions measures how concentrated the gains and losses are in time; twelve months represent about 13% to 14% of the available observations. Sensitivity checks use six and eighteen influential months, 18- and 24-month ranking windows, three descriptive subperiods, and exclusions based on the largest RW losses. No model is re-estimated for these checks.

6 Empirical Results

6.1 Average accuracy across targets and horizons

Table 2 averages performance across the sixteen target-horizon cells. XGBoost has the best mean RMSE rank, 4.00, and an average relative RMSE of 0.821, a 17.9% reduction from RW. Random Forest and Boruta

RF have similar average ratios of 0.825 and 0.826. The three combinations lie between 0.847 and 0.859 and rank near the top despite winning relatively few individual cells. Ridge, LASSO, and the factor model improve on RW on average but by smaller margins, while Elastic Net is slightly worse than RW. These are numerical rankings; the inferential results show much weaker separation.

Table 2. Average performance and inferential summary across 16 target-horizon cells

Model	Mean rank	Rel. RMSE	RMSE gain	Winner cells	MCS survival (SE)	DM wins (SE)
XGBoost	4.00	0.821	17.9%	5	100.0%	0
Mean Ensemble	4.50	0.847	15.3%	2	100.0%	0
Median Ensemble	4.62	0.859	14.1%	2	100.0%	0
Inverse-RMSE Ensemble	5.12	0.848	15.2%	0	100.0%	0
Random Forest	5.19	0.825	17.5%	1	87.5%	0
Boruta RF	5.25	0.826	17.4%	2	93.8%	0
AR(4)	6.50	0.869	13.1%	2	100.0%	0
Ridge	7.56	0.931	6.9%	1	100.0%	0
LASSO	7.81	0.943	5.7%	1	100.0%	0
Elastic Net	8.06	1.009	-0.9%	1	100.0%	0
Factor	9.44	0.956	4.4%	0	100.0%	0
RW	9.88	1.000	0.0%	0	100.0%	-

Notes: Relative RMSE is averaged across four targets and four horizons. DM wins count Holm-adjusted 5% rejections in favor of the model. The RW row has no pairwise DM count because it is the benchmark.

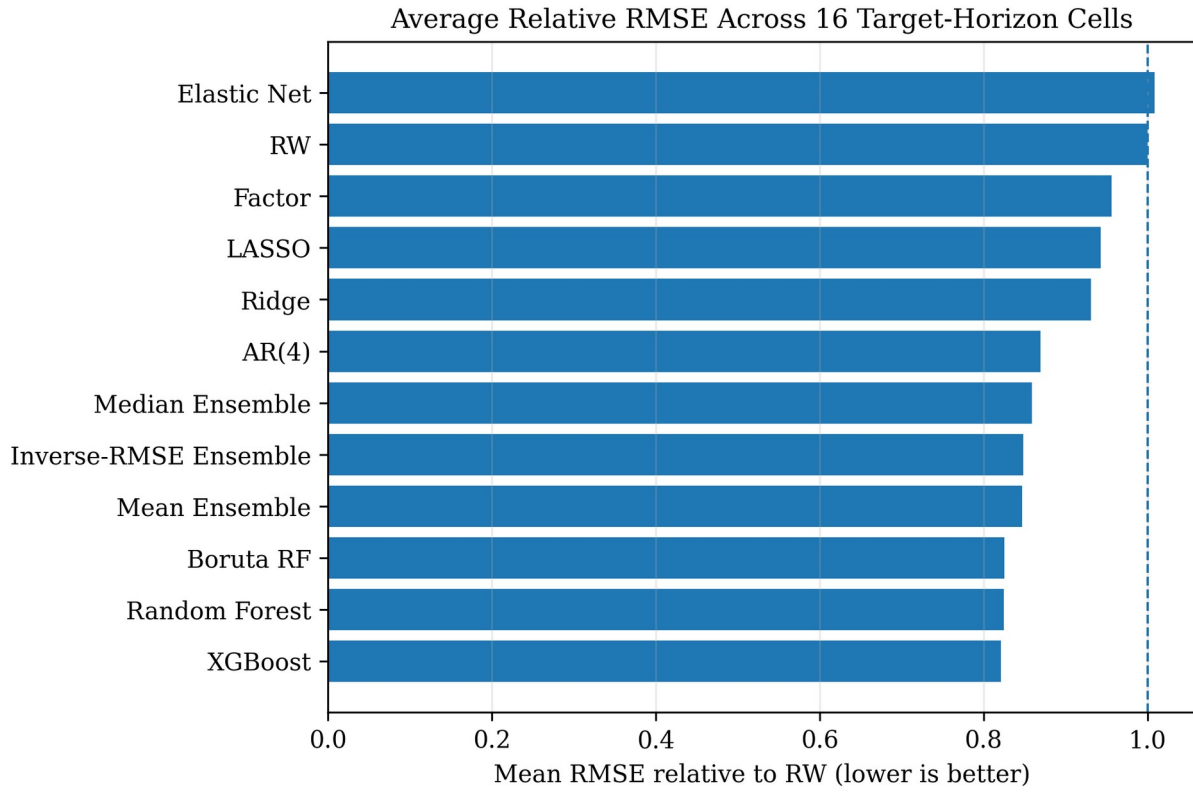


Figure 1. Average RMSE relative to transformed-series persistence

Notes: The dashed vertical line denotes equality with RW. The average is unweighted across the sixteen target-horizon cells.

6.2 Target and horizon heterogeneity

Performance varies materially by target and horizon. AR(4) leads one-month CPI and PCE inflation. XGBoost leads one- and three-month industrial production and the one-, three-, and six-month unemployment forecasts. Random Forest leads six-month CPI inflation, while Boruta RF leads three- and six-month PCE inflation. Forecast combinations lead several other cells. At the twelve-month industrial-production horizon, LASSO and Elastic Net generate the same intercept-only forecast and share the numerical winner. Ridge leads twelve-month unemployment. The breadth of winners is itself evidence against a universal ranking.

Table 3. RMSE winner by target and forecast horizon

Target	h	RMSE winner	Rel. RMSE	Gain	DM-SE	DM-AE
CPIAUCSL	1	AR(4)	0.917	8.3%	No	No
CPIAUCSL	3	Mean Ensemble	0.889	11.1%	No	No
CPIAUCSL	6	Random Forest	0.847	15.3%	No	No
CPIAUCSL	12	Median Ensemble	0.761	23.9%	No	No
INDPRO	1	XGBoost	0.783	21.7%	No	No
INDPRO	3	XGBoost	0.653	34.7%	No	No
INDPRO	6	Median Ensemble	0.721	27.9%	No	No
INDPRO	12	LASSO / Elastic Net	0.707	29.3%	No	No
PCEPI	1	AR(4)	0.934	6.6%	No	No
PCEPI	3	Boruta RF	0.895	10.5%	No	No
PCEPI	6	Boruta RF	0.872	12.8%	No	No

Target	h	RMSE winner	Rel. RMSE	Gain	DM-SE	DM-AE
PCEPI	12	Mean Ensemble	0.819	18.1%	No	No
UNRATE	1	XGBoost	0.772	22.8%	No	No
UNRATE	3	XGBoost	0.699	30.1%	No	No
UNRATE	6	XGBoost	0.694	30.6%	No	No
UNRATE	12	Ridge	0.712	28.8%	No	No

Notes: DM-SE and DM-AE indicate whether the winning model is significantly better than RW after Holm correction under squared or absolute loss. Numerical winner status does not imply statistical significance.

6.3 Statistical distinguishability

Formal comparisons are much less decisive than the RMSE rankings. Across 176 squared-error DM tests, no alternative is significantly more accurate than RW at the 5% level after Holm adjustment. Under absolute-error loss, five rejections remain, all for one-month industrial production: Boruta RF, Random Forest, and the three forecast combinations. No alternative is significantly worse than RW after adjustment under either loss.

The MCS results make the model uncertainty explicit. All twelve approaches survive in 25 of 32 target-horizon-loss panels; five panels retain eleven models and two retain ten. The mean survivor count is 11.72. RW therefore remains statistically non-eliminated in most panels even when it ranks poorly numerically. This is not evidence that the models are equally accurate; it is evidence that the available loss sequences do not support aggressive elimination at the chosen confidence level.

Table 4. Pairwise and multi-model inference by target and horizon

Target	h	DM wins SE	DM wins AE	MCS survivors SE	MCS survivors AE
CPIAUCSL	1	0	0	12/12	12/12
CPIAUCSL	3	0	0	12/12	12/12
CPIAUCSL	6	0	0	12/12	12/12
CPIAUCSL	12	0	0	11/12	10/12
INDPRO	1	0	5	12/12	11/12
INDPRO	3	0	0	12/12	12/12
INDPRO	6	0	0	11/12	12/12
INDPRO	12	0	0	11/12	10/12
PCEPI	1	0	0	12/12	12/12
PCEPI	3	0	0	12/12	12/12
PCEPI	6	0	0	12/12	12/12
PCEPI	12	0	0	12/12	12/12
UNRATE	1	0	0	12/12	11/12
UNRATE	3	0	0	12/12	12/12
UNRATE	6	0	0	12/12	12/12
UNRATE	12	0	0	12/12	12/12

Notes: DM wins count Holm-adjusted rejections in favor of alternatives relative to RW. MCS uses 90% confidence, T_{max} , and 5,000 block-bootstrap replications.

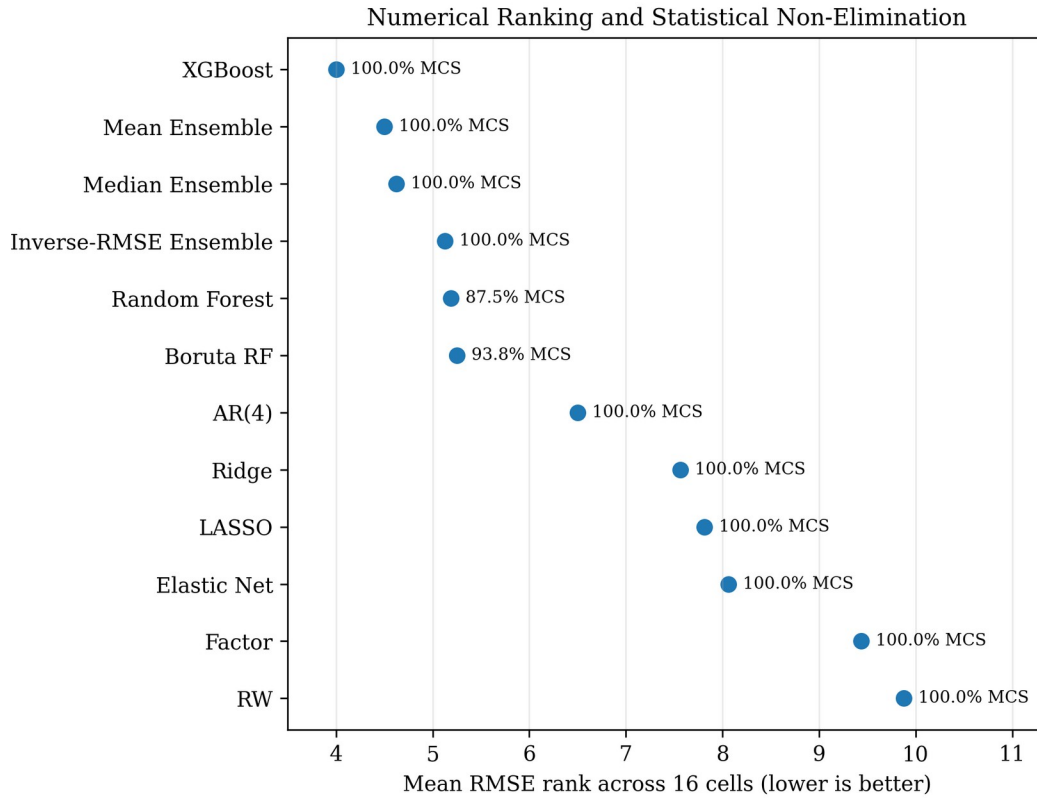


Figure 2. Numerical RMSE ranking versus MCS survival under squared-error loss

Notes: The figure illustrates that strong numerical ranking and statistical elimination are different concepts. XGBoost ranks first on average but, like most models, survives every squared-error MCS panel.

6.4 Winner instability and temporal concentration

Rolling 12-month rankings are unstable. Winner-set switch rates range from 17.9% for three-month PCE inflation to 38.5% for six-month PCE inflation. Each of the sixteen cells produces between seven and thirteen distinct winner sets. The mean switch rate is 29.1%, so a change in the leading set occurs in roughly three of every ten adjacent rolling comparisons.

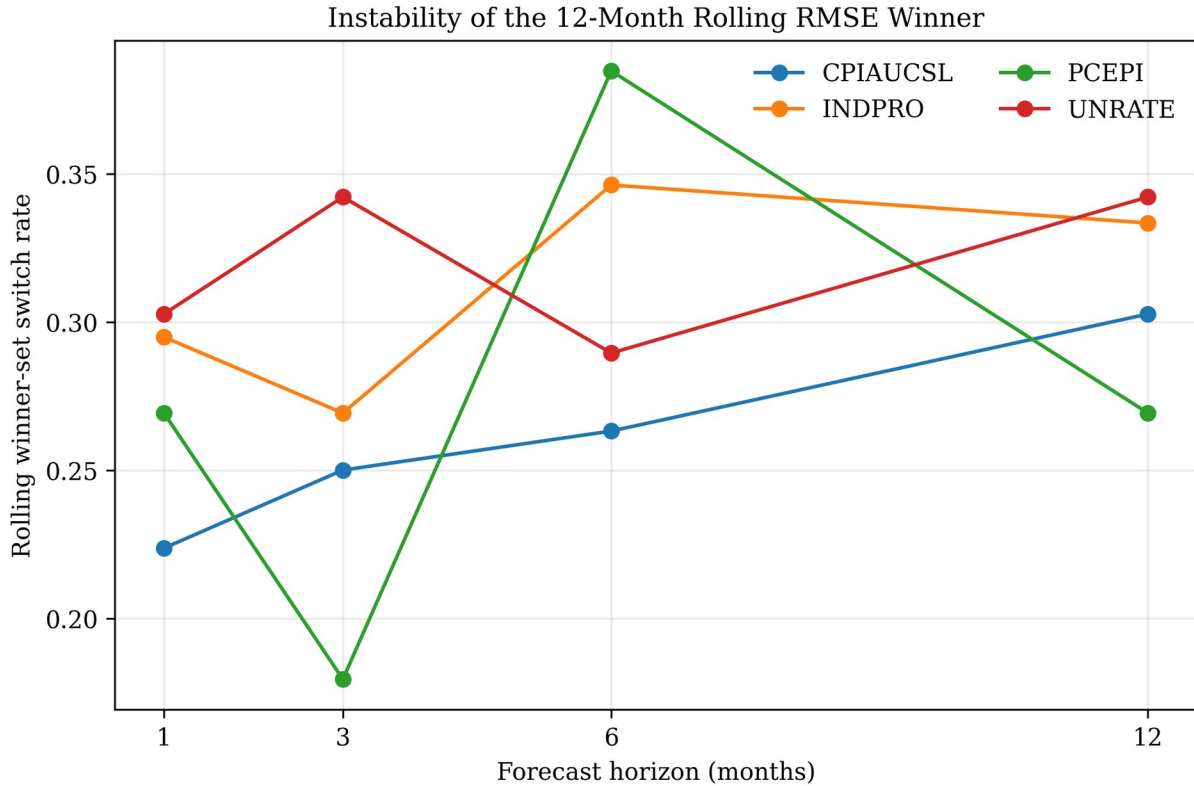


Figure 3. Switch rate of the 12-month rolling RMSE winner

Notes: The switch rate is winner-set changes divided by adjacent-window transitions. CPIAUCSL and UNRATE have 77 rolling windows because two transformed target months are excluded; PCEPI and INDPRO have 79.

Loss differences are concentrated in a small number of dates. The ranking is performed separately within each non-benchmark model-target-horizon cell. The twelve largest positive monthly loss reductions account for 85.6% of gross improvement on average, and the twelve largest deteriorations account for 89.1% of gross deterioration. These are not twelve common calendar months shared by all models. Rather, the calculation asks, cell by cell, how much of each model's total favorable and unfavorable performance is attributable to its most influential dates. A low average loss can therefore coexist with weak performance in many ordinary months.

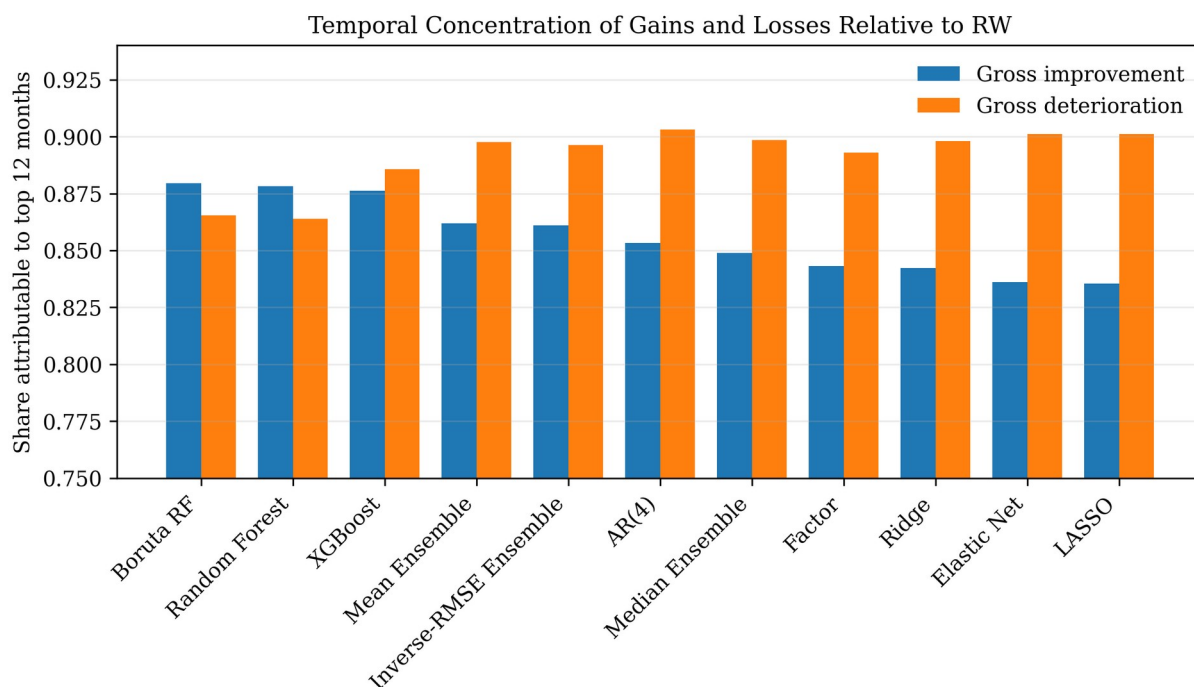


Figure 4. Temporal concentration of gross squared-loss gains and losses relative to RW

Notes: For each target-horizon-model cell, months are ranked separately by positive loss reduction and loss deterioration. The plotted values are averages of the shares attributable to the twelve largest months.

6.5 COVID and reopening sensitivity

The shock-window exercise removes target months from March 2020 through December 2021. The interval is fixed in advance as a descriptive COVID and reopening window, not estimated as a regime. Across model-horizon cells within each target, the median share of total squared loss generated during these 22 months is 94.1% for industrial production and 99.4% for unemployment, compared with 48.0% for CPI inflation and 44.1% for PCE inflation. The episode therefore dominates the real-activity and labor-market loss totals.

Removing the window changes the average ordering. The median, mean, and inverse-RMSE combinations become the three leading methods, with average relative RMSEs of about 0.764 to 0.766. AR(4), Ridge, Random Forest, and XGBoost follow. Boruta RF deteriorates more sharply than the other leading full-sample methods, indicating that part of its apparent advantage is specific to the shock period.

Table 5. Average relative RMSE with and without the COVID/reopening window

Model	Full OOS	Excl. shock window	Full rank	Excl. rank
Median Ensemble	0.859	0.764	6	1
Mean Ensemble	0.847	0.766	4	2
Inverse-RMSE Ensemble	0.848	0.766	5	3
AR(4)	0.869	0.779	7	4
Ridge	0.931	0.783	8	5
Random Forest	0.825	0.805	2	6
XGBoost	0.821	0.811	1	7
Factor	0.956	0.834	10	8
Elastic Net	1.009	0.841	12	9
LASSO	0.943	0.844	9	10
RW	1.000	1.000	11	11
Boruta RF	0.826	1.158	3	12

Notes: Ratios are first calculated within each target-horizon cell and then averaged across the sixteen cells. The exclusion is a descriptive sensitivity analysis and is not used to retune any model. Full and exclusion ranks order the average ratios reported in this table and therefore differ from the mean cell ranks in Table 2.

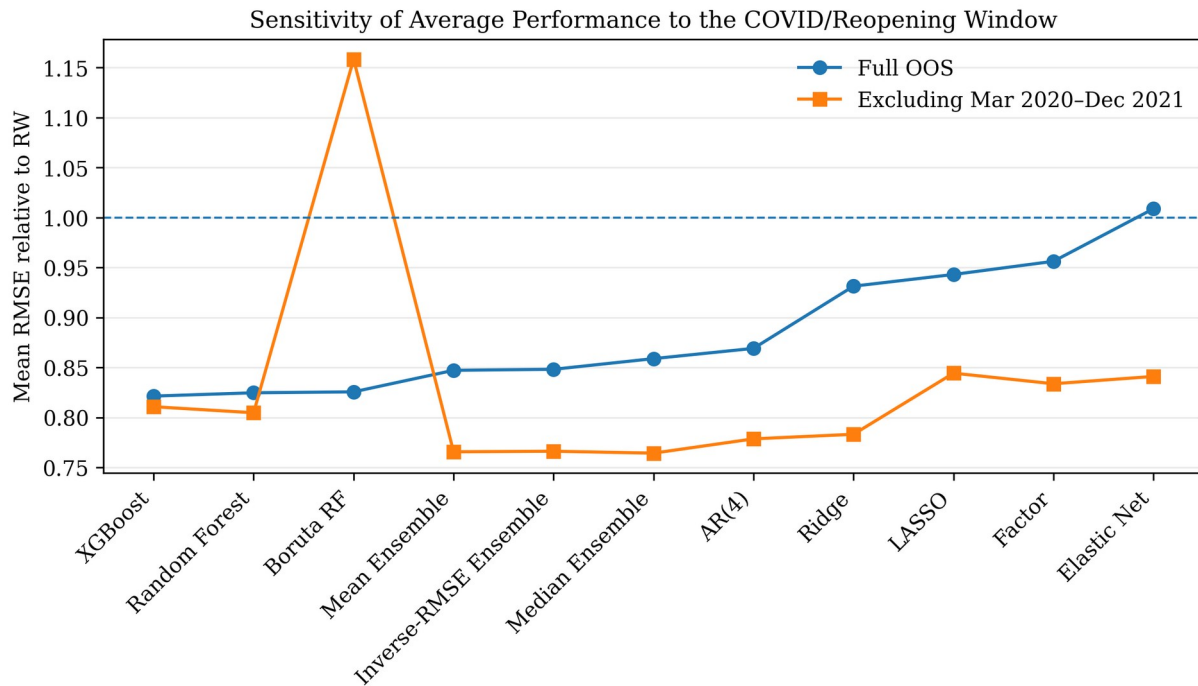


Figure 5. Sensitivity of average relative RMSE to excluding March 2020–December 2021

Notes: The ordering changes materially when the shock window is removed, reinforcing the distinction between average accuracy and temporal robustness.

6.6 Additional sensitivity checks

Three further checks use only the archived forecast panels; no model is re-estimated or retuned. A descriptive period split separates December 2018–February 2020, March 2020–December 2021, and January 2022–May 2026. Random Forest and the combinations lead before COVID, Boruta RF and XGBoost lead during the shock window, and the combinations lead from 2022 onward. The pre-COVID segment contains

only 15 target months, so the split is descriptive rather than an independent performance test. Even so, the leading model family changes by period.

Table 5A. Descriptive period decomposition of average relative RMSE

Period	Target months per cell	Three leading alternatives (mean relative RMSE)
Pre-COVID	15	Random Forest 0.743; Inverse-RMSE Ensemble 0.743; Mean Ensemble 0.743
COVID/reopening	22	Boruta RF 0.871; XGBoost 0.881; AR(4) 0.908
2022 onward	51-53	Median Ensemble 0.765; Mean Ensemble 0.768; Inverse-RMSE Ensemble 0.769

Notes: Relative RMSE is calculated within each target-horizon cell and then averaged across the sixteen cells. CPIAUCSL and UNRATE have two fewer evaluation months in the 2022-onward period because transformed observations affected by target imputation are excluded.

A benchmark-tail attribution exercise then removes, within each target-horizon cell, the dates with the largest RW squared errors and recomputes relative RMSE on the remaining observations. The same dates are removed for every model in the cell. Excluding only the largest benchmark-loss date raises the best average ratio from 0.821 to 0.916. After six exclusions, the best ratio is 0.988, and after twelve exclusions RW leads on average. This is deliberately a benchmark-centered attribution exercise, not a neutral outlier-deletion rule and not evidence that the removed observations are erroneous. It measures how much of the relative ranking is associated with the benchmark's most difficult dates.

The instability findings also survive alternative diagnostic choices. The six largest contributions account for 74.9% of gross improvements and 75.6% of gross deteriorations on average; the shares rise to 91.1% and 95.3% for eighteen contributions. Mean winner-set switch rates decline from 29.1% with 12-month windows to 24.1% and 20.4% with 18- and 24-month windows, but they remain economically meaningful.

Table 5B. Benchmark-tail attribution and diagnostic-window sensitivity

Panel A. Exclusion of largest RW-loss dates within each target-horizon cell			
Dates excluded per cell	Best average model	Best relative RMSE	Best alternative if RW leads
0	XGBoost	0.821	-
1	Inverse-RMSE Ensemble	0.916	-
3	Median Ensemble	0.950	-
6	Median Ensemble	0.988	-
12	RW	1.000	Median Ensemble 1.092
Panel B. Temporal concentration of gross squared-loss differences			
Top dates	Gross improvement share	Gross deterioration share	Model-cell observations
6	74.9%	75.6%	176
12	85.6%	89.1%	176
18	91.1%	95.3%	176
Panel C. Rolling RMSE winner-set turnover			
Window length	Mean switch rate	Cell-specific range	Mean distinct winner sets
12 months	29.1%	17.9% to 38.5%	9.6
18 months	24.1%	18.1% to 32.9%	8.2
24 months	20.4%	10.6% to 34.4%	6.6

Notes: Panel A removes dates according to the magnitude of the RW squared error, using the same excluded dates for every model within a target-horizon cell. Panel B averages across 176 non-benchmark model-target-horizon cells, whereas Panel C averages across the sixteen target-horizon cells. All calculations use the frozen forecasts and do not alter model estimation or tuning.

7 Implementation Diagnostics and Reproducibility

The computational release links each reported statistic to frozen forecast panels and validation records. All four target-specific experiments use the same data checksum, prespecified configuration, 360-month estimation window, 90-month OOS specification, model set, horizons, and MCS configuration with 5,000 bootstrap replications. Before target-specific evaluation exclusions, the archived panels contain 17,280 rows: four targets by 90 target months by four horizons by twelve models.

A separate recalculation from the archived panels reproduces RMSE, MAE, bias, correlation, relative metrics, DM statistics, Holm adjustments, and ensemble weights to numerical precision. Target transformations match direct calculations from the frozen source data. Target-origin date gaps equal the requested horizons, every estimation window contains exactly 360 months, and inverse-RMSE combinations use only outcomes observable by the current forecast origin.

Table 6. Diagnostics summary for the four target-specific experiments

Target	Rolling windows	Initial-missing windows	Max EM iterations	Boruta fallback	Raw target fills	Eval. exclusions
CPIAUCSL	360	274	0	0	1	2
PCEPI	360	274	0	0	0	0
INDPRO	360	274	0	0	0	0
UNRATE	360	274	0	1	1	2

Notes: Initial-missing windows contain at least one predictor missing value before eligibility screening and causal short-gap filling. Screening removed ineligible variables, and causal filling resolved the remaining eligible gaps; iterative PCA reconstruction was therefore not required in the frozen evaluation. The maximum EM iteration count was zero for all targets, no window was nonconverged, and no missing value remained after preprocessing. The single Boruta fallback occurs for UNRATE at $h=3$.

Two implementation details warrant disclosure. Some late-sample ragged-edge windows contain short-gap predictor fills, but those fills are causal and use only information available at the origin. The pipeline also produces a supplementary HAC-adjusted post-selection MAE comparison based on a Newey-West covariance estimate (Newey and West, 1987). This diagnostic is not a formal Giacomini-White conditional predictive ability test (Giacomini and White, 2006) and does not enter the primary claims.

8 Discussion and Limitations

A lower average RMSE is informative, but it is not a complete deployment rule. XGBoost lowers RMSE by 17.9% on average relative to RW, a numerically sizable gain; yet the absence of Holm-adjusted squared-error DM rejections and the broad MCS sets leave substantial uncertainty about whether it is uniquely superior. The leading model also changes with the target, horizon, and evaluation period. Rankings are therefore most useful when accompanied by inferential and temporal diagnostics.

Forecast combinations are a practical way to manage that uncertainty, although the results do not establish that a combination is uniformly superior. They rank well on average, avoid dependence on one algorithm, and become the leading group once the COVID/reopening window is removed. The median rule is less exposed to an isolated extreme forecast, while the mean and inverse-RMSE rules diversify model-specific errors in different ways. Their gains are also concentrated in time, so combining forecasts does not remove the need for sensitivity analysis.

The interpretation is subject to several limits. The experiment uses a current-vintage FRED-MD file rather than genuine real-time vintages, so historical revisions and release lags are not reproduced. The OOS

period covers at most 90 target months (88 for CPIAUCSL and UNRATE after exclusions) and includes an extraordinary shock, limiting both test power and the number of independent economic regimes. Tuning choices are deliberately restrained for transparency and comparability, and stochastic learners are evaluated with one prespecified origin-specific seed schedule rather than repeated seeds. The factor count is fixed at four for the factor model and the PCA fallback; because the fallback was not activated in the frozen evaluation, these results do not assess sensitivity to alternative fallback-imputation methods. Finally, the model universe excludes neural networks and foundation models. Their omission is a scope decision, not a claim of data incompatibility: credible inclusion would require a separate design for architecture, regularization, initialization, stochastic variation, external pretraining data, and vintage contamination. All conclusions are conditional on the stated data vintage, model universe, and preprocessing protocol.

A natural extension is a vintage-consistent evaluation over a longer OOS period. The same framework could then be used to study whether forecast disagreement and revision instability provide *ex ante* information about prediction risk, rather than being interpreted only after outcomes are observed.

9 Conclusion

Twelve forecasting approaches are compared across four U.S. macroeconomic targets and four horizons under a common rolling design. Several machine-learning methods achieve sizable numerical gains within this model universe: XGBoost ranks first on average, and random forests and forecast combinations also improve on transformed-series persistence. The statistical and temporal evidence is less decisive. No squared-error DM comparison survives Holm correction, most MCS panels retain every approach, rolling winners change frequently under 12-, 18-, and 24-month windows, and a small number of dates account for much of the gross gains and deteriorations.

Average accuracy, statistical distinguishability, and temporal durability are separate properties of forecast performance. Reporting only the first overstates what can be learned from a short and shock-heavy evaluation sample. In this application, the evidence supports using forecast combinations and uncertainty diagnostics alongside the leading models rather than treating one *ex post* winner as definitive.

Declarations

Data and Code Availability

The frozen research release is identified by data checksum 8591dd9f169f7aeb45b7c91782fbd947. A public replication package containing the research code, configuration, frozen output tables, validation summaries, and reproduction instructions will be deposited in a permanent repository when the release is finalized. Until then, the interactive research dashboard is available at https://slackquant.shinyapps.io/ml_econ_forecasting/.

Declaration of Generative AI and AI-Assisted Technologies

During preparation of this manuscript, the author used OpenAI ChatGPT for technical and editorial support, including code review, diagnostic checking, literature organization, and language and formatting assistance. The author independently executed the analysis, verified the empirical results and citations, made all research-design and interpretive decisions, and reviewed and revised the final text. The author takes full responsibility for the work.

Conflict of Interest

The author declares no conflicts of interest.

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Appendix A. Full Relative-RMSE Tables

Tables A1-A4 report RMSE relative to RW for every model, target, and horizon. Values below one indicate lower RMSE than transformed-series persistence. The main text reports inference separately because the numerical ratio alone does not establish a statistically significant gain.

Table A1. Relative RMSE for CPIAUCSL

Model	h=1	h=3	h=6	h=12
XGBoost	0.995	0.889	0.882	0.765
Mean Ensemble	0.944	0.889	0.850	0.767
Median Ensemble	0.936	0.925	0.878	0.761
Inverse-RMSE Ensemble	0.944	0.891	0.852	0.770
Random Forest	0.983	0.900	0.847	0.797
Boruta RF	0.976	0.904	0.860	0.818
AR(4)	0.917	0.895	0.889	0.773
Ridge	1.108	1.023	1.007	0.764
LASSO	0.967	0.929	0.917	0.764
Elastic Net	0.975	0.925	0.927	0.767
Factor	1.168	1.145	0.931	0.803
RW	1.000	1.000	1.000	1.000

Table A2. Relative RMSE for PCEPI

Model	h=1	h=3	h=6	h=12
XGBoost	1.010	1.042	0.983	0.826
Mean Ensemble	0.950	0.978	0.907	0.819
Median Ensemble	0.969	1.037	0.959	0.837
Inverse-RMSE Ensemble	0.945	0.973	0.906	0.823
Random Forest	1.011	0.984	0.913	0.844
Boruta RF	0.941	0.895	0.872	0.846
AR(4)	0.934	0.984	0.970	0.866
Ridge	1.197	1.152	1.067	0.845
LASSO	1.002	1.078	1.000	0.839
Elastic Net	1.022	1.077	1.000	0.839
Factor	0.983	1.227	0.971	0.919
RW	1.000	1.000	1.000	1.000

Table A3. Relative RMSE for INDPRO

Model	h=1	h=3	h=6	h=12
XGBoost	0.783	0.653	0.731	0.707
Mean Ensemble	0.955	0.728	0.740	0.713
Median Ensemble	1.027	0.728	0.721	0.707
Inverse-RMSE Ensemble	0.966	0.729	0.741	0.713
Random Forest	0.836	0.692	0.726	0.723
Boruta RF	0.798	0.694	0.760	0.769
AR(4)	0.946	0.742	0.744	0.709
Ridge	1.210	0.699	0.737	0.707
LASSO	1.100	0.734	0.888	0.707
Elastic Net	1.094	0.728	1.181	0.707
Factor	1.345	0.833	0.779	0.738
RW	1.000	1.000	1.000	1.000

Table A4. Relative RMSE for UNRATE

Model	h=1	h=3	h=6	h=12
XGBoost	0.772	0.699	0.694	0.713
Mean Ensemble	0.987	0.834	0.775	0.720
Median Ensemble	1.095	0.728	0.723	0.712
Inverse-RMSE Ensemble	0.988	0.834	0.775	0.720
Random Forest	0.782	0.735	0.698	0.723
Boruta RF	0.788	0.714	0.701	0.874
AR(4)	1.199	0.794	0.783	0.760
Ridge	1.235	0.722	0.718	0.712
LASSO	1.124	1.420	0.906	0.714
Elastic Net	1.155	1.389	1.643	0.713
Factor	1.197	0.763	0.785	0.713
RW	1.000	1.000	1.000	1.000

Appendix B. Reproducibility Protocol

The computational release was frozen only after all four target-specific experiments passed a common validation protocol. Table B1 summarizes the target-specific release coverage and validation status. The frozen source data, experiment manifests, validation summaries, and archived output inventory are linked by a shared MD5 checksum. Any future revision of the empirical results will be tied to a newly identified computational release.

Table B1. Computational release summary

Target	OOS start	OOS end	Window	Validation	Data MD5
CPIAUCSL	2018-12-01	2026-05-01	360	Passed	8591dd9f16...
PCEPI	2018-12-01	2026-05-01	360	Passed	8591dd9f16...
INDPRO	2018-12-01	2026-05-01	360	Passed	8591dd9f16...
UNRATE	2018-12-01	2026-05-01	360	Passed	8591dd9f16...

Notes: The complete checksum is 8591dd9f169f7aeb45b7c91782fbd947.

The computational environment was R 4.6.0 on Windows 11. Key package versions include glmnet 5.0, randomForest 4.7-1.2, xgboost 3.2.1.1, Boruta 10.0.0, forecast 9.0.2, MCS 0.2.0, sandwich 3.1-2, dplyr 1.2.1, and ggplot2 4.0.3. Complete session information is retained in the frozen release.

The archived configuration records the effective transformation code for every target and predictor, the origin-specific active predictor set, short-gap fills, PCA-fallback diagnostics, the factor count, the use of the origin value plus three monthly lags in the maximum design, the time-ordered validation rule, stochastic seeds, ensemble weights, and all DM and MCS settings. The public package will replace local absolute paths with relative paths while preserving the internal frozen release unchanged.